NYCU IMF - 114 Financial Econometrics

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Assignment: 4, Ch5 q5, 6, 20

Question 5

For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|x)$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T\to\infty} \text{var}(b_2|x) = 0$.

(a) $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \ldots, T$. Note that $x = (1, 2, \ldots, T), \Sigma_{t=1}^T (t - \bar{t})^2 = \Sigma_{t=1}^T t^2 - (\Sigma_{t=1}^T t)^2/T, \Sigma_{t=1}^T t = T(T+1)/2$ and $\Sigma_{t=1}^T t^2 = T(T+1)(2T+1)/6$

Linear Trend: $y_t = \beta_1 + \beta_2 t + e_t$

Since MR1–MR6 hold, the conditional variance of the OLS estimator for a simple linear regression is:

$$\text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\sum(x_t - \bar{x})^2}$$

First, let's calculate the denominator $\sum_{t=1}^T (t - \bar{t})^2$. Using the hint provided:

1. $\sum t = \frac{T(T+1)}{2}$
2. $\sum t^2 = \frac{T(T+1)(2T+1)}{6}$
3. $(\sum t)^2/T = \frac{T^2(T+1)^2}{4T} = \frac{T(T+1)^2}{4}$

Now, find the sum of squares:

$$\sum(t - \bar{t})^2 = \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4}$$

Factoring out $\frac{T(T+1)}{2}$:

$$\begin{aligned} &= \frac{T(T+1)}{2}\left[\frac{2T+1}{3} - \frac{T+1}{2}\right] \\ &= \frac{T(T+1)}{2}\left[\frac{4T+2-3T-3}{6}\right] \\ &= \frac{T(T+1)(T-1)}{12} \\ &= \frac{T(T^2-1)}{12} \end{aligned}$$

The variance is:

$$\text{var}(b_2|\mathbf{x}) = \frac{12\sigma^2}{T(T^2-1)}$$

Check for consistency:

$$\lim_{T\to\infty} \text{var}(b_2|\mathbf{x}) = \lim_{T\to\infty} \frac{12\sigma^2}{T^3 - T} = 0$$

Since the variance collapses to zero as the sample size increases, the estimator b_2 is consistent.

(b) $y_t = \beta_1 + \beta_2(0.5)^t + e_t, t = 1, 2, \ldots, T$. Here, $x = (0.5, 0.5^2, \ldots, 0.5^T)$. Note that the sum of a geometric progression with first term and common ratio is

$$S = r + r^2 + r^3 + \cdots + r^n = \frac{r(1 - r^n)}{1 - r}$$

Geometric Trend: $y_t = \beta_1 + \beta_2(0.5)^t + e_t$
Here, $x_t = r^t$ where $r = 0.5$. To simplify, we look at the denominator $\sum(x_t - \bar{x})^2$. As $T \to \infty$, the behavior of this sum is dominated by whether $\sum x_t^2$ converges.

Using the geometric series formula for $x_t^2 = (0.5^2)^t = (0.25)^t$:

$$\sum_{t=1}^{T}(0.25)^t = \frac{0.25(1 - 0.25^T)}{1 - 0.25}$$

As $T \to \infty$, this sum converges to a finite constant: $\frac{0.25}{0.75} = \frac{1}{3}$.
Since the sum of squares in the denominator approaches a fixed finite number rather than infinity:

$$\lim_{T\to\infty} \text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\text{finite value}} \neq 0$$

The estimator b_2 is not consistent.

(c) Provide an intuitive explanation for these results.

The difference in consistency comes down to how much "new information" each additional observation provides about the parameter β_2.

- In (a): The regressor t grows larger and larger. As T increases, the "spread" of the independent variable ($\sum(t - \bar{t})^2$) grows at a rate of T^3. This huge variation in x makes it very easy for OLS to pin down the slope β_2 with extreme precision, eventually reaching 0 variance.
- In (b): The regressor $(0.5)^t$ shrinks toward zero very quickly. After just a few observations (e.g., $0.5, 0.25, 0.125...$), the subsequent values of x_t are almost zero. Effectively, the model stops receiving "new" information about the slope because the "signal" in x has vanished. Because the total variation in x is limited (bounded), the uncertainty in β_2 remains, no matter how many more data points you collect.

Question 6

Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

(a) $\beta_2 = 0$

Preliminary Information

- Sample size (N): 63
- Number of parameters (K): 3 ($\beta_1, \beta_2, \beta_3$)
- Degrees of freedom (df): $N - K = 63 - 3 = 60$
- Significance level: 5% (two-tailed test)
- Critical value: $t_c = t_{(0.975,60)} \approx 2.0003$

Test $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$

1. Estimate: $b_2 = 3$
2. Variance: From the covariance matrix, $\widehat{\text{var}}(b_2)$ is the element in the 2nd row, 2nd column: 4.
3. Standard Error: $se(b_2) = \sqrt{4} = 2$.
4. Test Statistic:
$$t = \frac{3 - 0}{2} = 1.5$$
5. Conclusion: Since $|1.5| < 2.0003$, we fail to reject the null hypothesis. β_2 is not significantly different from zero at the 5% level.

(b) $\beta_1 + 2\beta_2 = 5$

Test $H_0 : \beta_1 + 2\beta_2 = 5$ against $H_1 : \beta_1 + 2\beta_2 \neq 5$

1. Estimated Value: $L = b_1 + 2b_2 = 2 + 2(3) = 8$
2. Variance of the combination:
$$\widehat{\text{var}}(b_1 + 2b_2) = \widehat{\text{var}}(b_1) + 4\widehat{\text{var}}(b_2) + 4\widehat{\text{cov}}(b_1, b_2)$$
Using the matrix values: $3 + 4(4) + 4(-2) = 3 + 16 - 8 = 11$.
3. Standard Error: $se(L) = \sqrt{11} \approx 3.3166$
4. Test Statistic:
$$t = \frac{8 - 5}{3.3166} = \frac{3}{3.3166} \approx 0.9045$$

5. Conclusion: Since $|0.9045| < 2.0003$, we fail to reject the null hypothesis.

(c) $\beta_1 - \beta_2 + \beta_3 = 4$
Test $H_0 : \beta_1 - \beta_2 + \beta_3 = 4$ against $H_1 : \beta_1 - \beta_2 + \beta_3 \neq 4$

1. Estimated Value: $L = 2 - 3 + (-1) = -2$
2. Variance of the combination:For a combination $L = a_1 b_1 + a_2 b_2 + a_3 b_3$, the variance is $\mathbf{a}'\widehat{\text{cov}}(b)\mathbf{a}$.

$$\widehat{\text{var}}(L) = \text{var}(b_1) + \text{var}(b_2) + \text{var}(b_3) - 2\text{cov}(b_1, b_2) + 2\text{cov}(b_1, b_3) - 2\text{cov}(b_2, b_3)$$

Plugging in the numbers:

$$= 3 + 4 + 3 - 2(-2) + 2(1) - 2(0)$$

$$= 10 + 4 + 2 = 16$$

3. Standard Error: $se(L) = \sqrt{16} = 4$
4. Test Statistic:
$$t = \frac{-2 - 4}{4} = \frac{-6}{4} = -1.5$$
5. Conclusion: Since $|-1.5| < 2.000$, we fail to reject the null hypothesis.

Question 20
A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \ldots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y_2} - \bar{y_1}}{\bar{x_2} - \bar{x_1}}$$

where $(\bar{y_1}, \bar{x_1})$ and $(\bar{y_2}, \bar{x_2})$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

(a) Show that $\hat{\beta}_{2,mean}$ is unbiased.
Substitute $y_i = \beta_1 + \beta_2 x_i + e_i$ into the means for each group:

$$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1 \quad \text{and} \quad \bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

Substitute these into the estimator formula:

$$\begin{aligned}
\hat{\beta}_{2,mean} &= \frac{(\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \\
&= \frac{\beta_2(\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \\
&= \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}
\end{aligned}$$

Taking the expectation conditional on x:

$$
\begin{aligned}
E(\hat{\beta}_{2,mean}|\mathbf{x}) &= \beta_2 + \frac{E(\bar{e}_2) - E(\bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \\
&= \beta_2 + 0 \\
&= \beta_2
\end{aligned}
$$

Thus, the estimator is unbiased.

(b) Derive an expression for $\text{var}(\hat{\beta}_{2,mean}|x)$.
From part a, we established the following expression for the estimator:

$$\hat{\beta}_{2,mean} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

When we take the variance conditional on $\mathbf{x}$, the parameters β_2 and the values of x are treated as constants. The only source of randomness is the error terms:

$$\text{var}(\hat{\beta}_{2,mean}|\mathbf{x}) = \text{var}\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right)$$

1. Remove constants: The variance of a constant (β_2) is zero, and the denominator $(\bar{x}_2 - \bar{x}_1)$ is a constant that comes out squared:

$$\text{var}(\hat{\beta}_{2,mean}|\mathbf{x}) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\text{var}(\bar{e}_2 - \bar{e}_1)$$

2. Independence of Groups: Since the first half and second half of the sample are independent (MR4), the variance of the difference is the sum of the variances:

$$\text{var}(\bar{e}_2 - \bar{e}_1) = \text{var}(\bar{e}_2) + \text{var}(\bar{e}_1)$$

3. Variance of a Sample Mean: For each group, the sample size is $N/2$. The variance of the mean of n i.i.d. variables is σ^2/n. Here, $n = N/2$:

$$\text{var}(\bar{e}_1) = \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N} \quad \text{and} \quad \text{var}(\bar{e}_2) = \frac{2\sigma^2}{N}$$

4. Final Substitution:

$$\text{var}(\hat{\beta}_{2,mean}|\mathbf{x}) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\left(\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}\right) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

(c) Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
Using the Law of Total Variance:

$$\text{var}(\hat{\beta}_{2,mean}) = E[\text{var}(\hat{\beta}_{2,mean}|\mathbf{x})] + \text{var}(E[\hat{\beta}_{2,mean}|\mathbf{x}])$$

1. Second Term: We know from part a that $E[\hat{\beta}_{2,mean}|\mathbf{x}] = \beta_2$. Since β_2 is a constant, its variance is 0.
2. First Term: We plug in the result from part b:

$$\text{var}(\hat{\beta}_{2,mean}) = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right]$$

Thus, the expression is:

$$\text{var}(\hat{\beta}_{2,mean}) = \frac{4\sigma^2}{N}E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$$

(d) Under what conditions will $\hat{\beta}_{2,mean}$ var be a consistent estimator for β_2?.
For an estimator to be consistent, it must converge in probability to the true parameter value as the sample size N approaches infinity. A standard way to prove consistency is to show two things:

1. The estimator is unbiased (or asymptotically unbiased).
2. The variance of the estimator collapses to zero as $N \to \infty$.

In part (a) We already proved that $E[\hat{\beta}_{2,mean}] = \beta_2$, so the estimator is unbiased. Therefore, consistency hinges entirely on the variance collapsing to zero.
The variance derived in part (b):

$$\text{var}(\hat{\beta}_{2,mean}|\mathbf{x}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

For this variance to approach 0 as $N \to \infty$, the denominator must approach infinity. Specifically, we need:

$$\lim_{N\to\infty} N(\bar{x}_2 - \bar{x}_1)^2 = \infty$$

Since N is growing to infinity, the only thing that could prevent the denominator from going to infinity is if the squared difference between the group means, $(\bar{x}_2 - \bar{x}_1)^2$, shrinks to zero at the same rate (or faster) than N grows.
Therefore, the critical condition for consistency is:

$$\text{plim}(\bar{x}_2 - \bar{x}_1) = c \neq 0$$

(where c is a constant)
Conclusion:
The estimator $\hat{\beta}_{2,mean}$ is consistent under the condition that the explanatory variable x has non-zero variance in the population, and the data is ordered/grouped such that the distance between the two group means $(\bar{x}_2 - \bar{x}_1)$ converges to a non-zero constant as $N \to \infty$.

(e) Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes N = 100,500,1000, and, if your software permits, N = 5000. Assuming $\sigma^2 = 1000$, for each sample size, compute:

i. $\text{var}(b_2|x)$ and $\text{var}(\hat{\beta}_{2,mean}|x)$ where b_2 is the OLS estimator.

ii. Estimates for $E[(S_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)2/12$.]

N	$\text{var}(b_2\|\mathbf{x})$	$\text{var}(\hat{\beta}_{2,mean}\|\mathbf{x})$	$E[(s_x^2)^{-1}]$	$E[4/(\bar{x}_2 - \bar{x}_1)^2]$
100	1.2436	1.6632	0.1244	0.1663
500	0.2440	0.3260	0.1220	0.1630
1000	0.1229	0.1656	0.1229	0.1656
5000	0.0241	0.0323	0.1207	0.1616

Table 1: Simulation Results comparing OLS and Grouped Means Estimator variances.

(f) Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that is $\hat{\beta}_{2,mean}$ consistent?
1. Relative Magnitudes [Part (e)(i)]
For every sample size N, the variance of the grouped means estimator $\text{var}(\hat{\beta}_{2,mean}|\mathbf{x})$ is strictly larger than the OLS estimator variance $\text{var}(b_2|\mathbf{x})$.

- For N = 100: $1.663 > 1.243$
- For N = 5000: $0.032 > 0.024$

This confirms the Gauss-Markov theorem: OLS is the more efficient estimator because it utilizes the exact x-distance of every single observation, while the grouped means estimator "loses" information by compressing the data into just two averages.

2. How they change as sample size increases [Part (e)(i) & (ii)]
As N increases from 100 to 5000, both $\text{var}(b_2|\mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean}|\mathbf{x})$ steadily decrease toward 0.
Meanwhile, the terms in part (ii) ($E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$) do not approach 0. They are stabilizing around specific constant values. $E[(s_x^2)^{-1}]$ is hovering around 0.12, and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ is hovering around 0.16.

3. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
Yes. Because the variance of $\hat{\beta}_{2,mean}$ shrinks toward 0 as the sample size increases (going from 1.663 down to 0.032), the estimator is consistent. As we proved in part d, this happens because the difference between the top and bottom groups ($E[4/(\bar{x}_2 - \bar{x}_1)^2]$) stabilizes as a constant, allowing the growing N in the denominator to drive the overall variance to zero.

(g) Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval

$$(a, b)$$

is (b-a)2/12.]
Mathematical proof using the hint provided in the text:
1. The Uniform Distribution Variance:
For a uniform random variable on the interval (a, b), the population variance is:

$$\sigma_x^2 = \frac{(b-a)^2}{12}$$

Since the data is drawn from a uniform distribution between 0 and 10:

$$\sigma_x^2 = \frac{(10-0)^2}{12} = \frac{100}{12} \approx 8.333$$

Therefore, the expected value of its inverse is:

$$\frac{1}{\sigma_x^2} = \frac{12}{100} = 0.12$$

2. The Grouped Means Difference:
When we split a Uniform(0, 10) distribution in half, we get two new uniform distributions:

- The bottom half is Uniform(0, 5). Its expected mean is exactly in the middle: 2.5
- The top half is Uniform(5, 10). Its expected mean is exactly in the middle: 7.5

So, as N approaches infinity, $\bar{x}_1$ converges to 2.5 and $\bar{x}_2$ converges to 7.5. Plugging this into the term from part (ii):

$$\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4}{(7.5-2.5)^2} = \frac{4}{5^2} = \frac{4}{25} = 0.16$$

(h) Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

No, it would absolutely not be consistent.
Reason:
1. The Behavior of the Means (Law of Large Numbers)
If we randomly assign the data into two halves without sorting by x, then Group 1 and Group 2 are effectively just two random samples drawn from the exact same underlying population distribution.
By the Law of Large Numbers, as the sample size N goes to infinity, the sample mean of a random group converges to the true population mean, $E[x]$.

- $\bar{x}_1 \xrightarrow{p} E[x]$
- $\bar{x}_2 \xrightarrow{p} E[x]$

2. The Collapse of the Denominator
Because both group means converge to the exact same value, the difference between them disappears:

$$\text{plim}(\bar{x}_2 - \bar{x}_1) = E[x] - E[x] = 0$$

If we look back at the formula for the estimator itself:

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

As N gets large, you are essentially trying to divide by zero. The estimator becomes completely unstable.

3. The Explosion of the Variance
We derived the variance in part (b) as:

$$\text{var}(\hat{\beta}_{2,mean}|\mathbf{x}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

For consistency, we need this variance to shrink to 0 as $N \to \infty$. However, if the data is randomly ordered, the term $(\bar{x}_2 - \bar{x}_1)^2$ shrinks toward 0 at the exact same time N is growing to infinity. Because the denominator is collapsing to zero, the variance will not collapse to zero (in fact, it blows up toward infinity).

Summary
The entire logic of this "grouped means" estimator relies on comparing a "low x" group to a "high x" group to calculate a slope $\left(\frac{\Delta y}{\Delta x}\right)$. If we randomize the groups, both groups have the exact same average x and the exact same average y. We lose the horizontal spread (Δx) entirely, leaving us with a vertical line where we cannot calculate a slope. Therefore, ordering the data is mathematically mandatory for this estimator to work.